PROJECT REPORT

On

SIMPLE DIGIT RECOGNIZER USING 5x5 GRID

(A Python Based Pattern Recognition System)

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Semester: Winter Semester 2025-2026

1. Introduction

Pattern recognition is an important concept in computer science and artificial intelligence. It involves identifying patterns and making decisions based on input data. In this project, a simple digit recognition system is developed using Python.

The system allows a user to draw a digit on a 5×5 grid using binary values (0 and 1). The program then compares the input with predefined templates and identifies the closest matching digit.

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2. Objective

The main objectives of this project are:

- To understand basic pattern recognition techniques
- To implement template matching using Python
- To develop a simple digit recognition system
- To analyze how similarity between patterns can be measured

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3. Problem Statement

The goal is to design a program that can recognize digits based on a userdefined 5×5 grid input. The system should compare the input with stored templates and determine the most similar digit.

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4. Methodology

The project uses a simple **template matching approach**:

1. Predefine templates for digits (0, 1, and 2) using 5×5 grids
2. Take input from the user in the form of a 5×5 grid
3. Compare the input grid with each template
4. Count matching and mismatching pixels
5. Calculate a matching score
6. Select the digit with the highest score
7. If the match is above a threshold (80%), display the result

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5. Algorithm

1. Start the program
2. Define templates for digits (0, 1, 2)
3. Take 5×5 grid input from the user
4. For each template: ○ Compare each cell with user input ○ Count mismatches ○ Calculate match score
5. Find the template with the highest score
6. If score ≥ 80%:
   - Display recognized digit Else: ○ Display "not recognized" message
7. Ask user to continue or exit
8. Stop

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6. Implementation Details • **Programming Language:** Python

- **Concept Used:** Template Matching

. Grid Size: 5×5

- **Input Type:** Binary values (0 and 1)
- **Output Type:** Recognized digit or error message

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7. Code Description

The program consists of the following main parts:

- **Templates Dictionary:** Stores predefined digit patterns
- **Input Function:** Takes user input row by row
- **Display Function:** Prints the grid using symbols
- **Recognition Function:** Compares input with templates and calculates match score

- **Main Loop:** Allows repeated execution

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8. Sample Input and Output Input:

0 1 1 1 0
1 0 0 0 1
1 0 0 0 1
1 0 0 0 1
0 1 1 1 0

Output:

The system recognized this as the digit: 0

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9. Features

- Easy to understand logic
- Interactive user input
- Displays matching score
- Simple implementation using basic Python

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10. Limitations

- Recognizes only digits 0, 1, and 2
- Fixed grid size (5×5)
- Not suitable for complex or noisy inputs
- Accuracy depends on input similarity

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11. Applications

- Basic learning of pattern recognition
- Educational demonstration of AI concepts
- Foundation for image processing systems

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12. Future Enhancements

- Add support for digits 0–9
- Improve matching algorithm
- Use machine learning techniques
- Develop graphical user interface (GUI)
- Allow larger and flexible grids

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13. Conclusion

This project demonstrates a simple approach to digit recognition using template matching. It helps in understanding how computers can identify patterns based on similarity. Although basic, it forms a foundation for more advanced systems in artificial intelligence and image processing.